

COMP5318 Assignment 2 Report: Image Classification

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COMP5318 Machine Learning and Data Mining

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Abstract—This report investigates image classification on the PathMNIST dataset using three machine learning algorithms: Logistic Regression, a multilayer perceptron (MLP), and a convolutional neural network (CNN). The study compares these models in terms of classification accuracy, F1-score, runtime, parameter complexity, and suitability for biomedical image classification. The report includes data exploration, preprocessing, model architecture design, hyperparameter tuning, and final test-set evaluation. Experimental results show that the CNN achieved the strongest overall performance because convolutional layers preserved local spatial image structure, while Logistic Regression provided a simple but limited linear baseline. The MLP achieved intermediate performance by learning nonlinear decision boundaries on flattened image representations.

Index Terms—PathMNIST, image classification, convolutional neural network, multilayer perceptron, Logistic Regression, hyperparameter tuning

I. INTRODUCTION

Medical image classification has become an increasingly important application of machine learning because accurate automated diagnosis can assist clinicians in analysing large volumes of biomedical data efficiently and consistently. Histopathology image analysis is particularly challenging due to the subtle visual differences between tissue types, variations in staining, and the complex local texture structures present in microscopic images. As a result, image classification models must be able to extract meaningful visual patterns while remaining robust to noise and intra-class variation.

This assignment investigates image classification on the PathMNIST dataset, a biomedical image benchmark consisting of small RGB histopathology image patches belonging to nine tissue categories. The objective is to compare the effectiveness of three different machine learning approaches for this task: Logistic Regression, a multilayer perceptron (MLP), and a convolutional neural network (CNN). These models represent increasing levels of modelling complexity, ranging from a simple linear classifier to deep neural architectures specifically designed for image processing.

Comparing multiple classifiers is important because different algorithms make different assumptions about the structure of the data and therefore exhibit different strengths and limitations. Logistic Regression provides a lightweight and interpretable baseline but may struggle with complex nonlinear image features. The MLP introduces nonlinear representation learning through hidden layers, while the CNN explicitly preserves spatial relationships between neighbouring pixels

TABLE I
SUMMARY OF DATASET SPLITS

Dataset Split	Number of Images	Image Size	Classes
Training Set	32,000	28×28	9
Test Set	8,000	28×28	9

using convolutional operations. Evaluating these approaches side-by-side allows analysis of trade-offs involving classification performance, overfitting behaviour, runtime, parameter complexity, and interpretability.

In addition to implementing the three models, this study also investigates the impact of hyperparameter tuning on predictive performance. Validation-based grid search was used to explore different architectural and optimisation settings for the neural network models. The final comparison is based on test-set accuracy, F1-score, confusion matrices, runtime, and model complexity. Through this analysis, the report aims to identify which modelling approach is most suitable for biomedical image classification under the constraints of the PathMNIST dataset.

II. DATA

A. Dataset Description

The dataset used in this study is PathMNIST from the MedMNIST benchmark collection, which is designed for lightweight biomedical image classification experiments. Each sample is a colour histopathology image patch of size $28 \times 28 \times 3$, belonging to one of nine tissue categories.

Although the dataset is relatively small compared with large-scale natural image datasets, the classification task remains challenging because many tissue categories share highly similar colour distributions and differ mainly in subtle local texture patterns and cellular arrangements. Furthermore, the low image resolution limits the amount of fine-grained structural information available to the models.

To support hyperparameter tuning and model selection, the original training set was further divided into a new training subset and validation subset using a stratified 80:20 split. Stratification preserves the class proportions across subsets and reduces the risk of biased validation performance estimates.

B. Data Exploration

Figure 1 shows representative examples from each PathMNIST class.

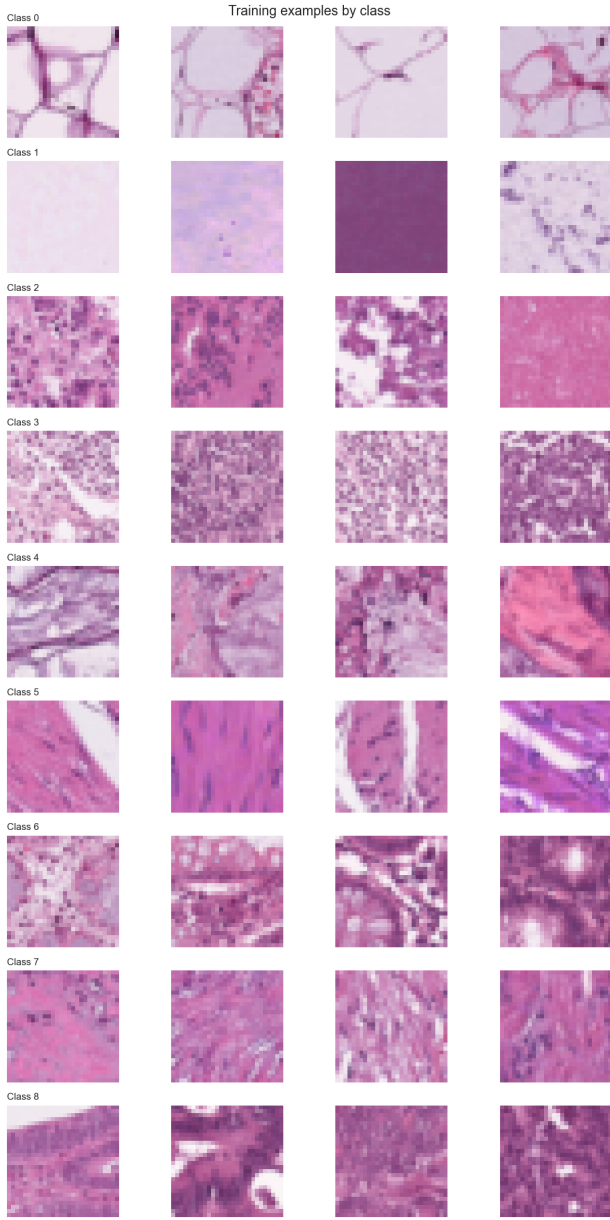


Fig. 1. Sample images from each PathMNIST class.

Several visual patterns can be observed from the sample images. Many classes exhibit similar staining colours dominated by pink and purple tones, meaning that simple global colour statistics alone are unlikely to separate the classes effectively. Instead, discrimination depends more strongly on local spatial texture, tissue organisation, and microscopic structural details. This characteristic suggests that models capable of preserving spatial relationships between neighbouring pixels, such as CNNs, may perform better than models relying on flattened feature vectors.

Figure 2 shows the class distribution in the training set.

The dataset is relatively balanced across the nine classes, although small differences in class frequency still exist. Because severe class imbalance is not present, overall accuracy remains

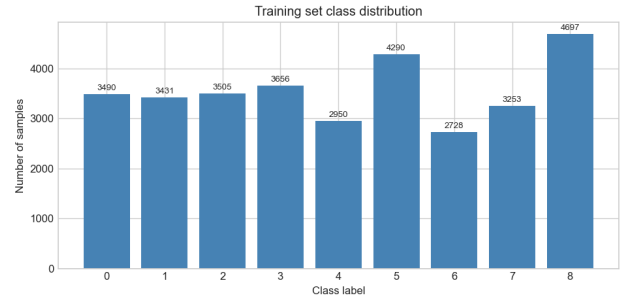


Fig. 2. Class distribution in the training set.

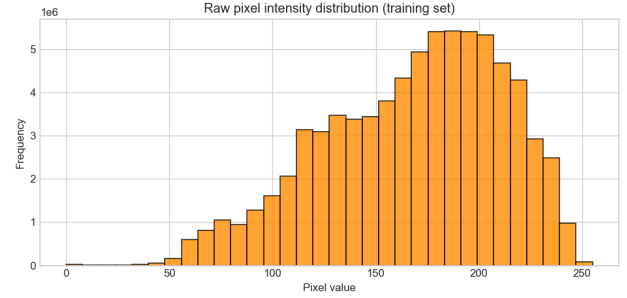


Fig. 3. Pixel value distribution before normalisation.

an informative evaluation metric. However, macro F1-score is also reported to ensure that model performance is evaluated consistently across all classes rather than being dominated by larger categories.

Figure 3 illustrates the distribution of pixel intensities before preprocessing.

The raw pixel values are concentrated within the standard 8-bit image range $[0, 255]$. Without preprocessing, large numerical scales may slow optimisation and produce unstable gradient updates during neural network training.

C. Preprocessing

Several preprocessing steps were applied before model training to improve numerical stability and ensure compatibility with different model architectures.

First, image arrays were converted from `uint8` to `float32`, and pixel values were normalised from the original range $[0, 255]$ to $[0, 1]$. This scaling reduces the magnitude of input values and improves the stability of gradient-based optimisation methods such as Adam.

Second, two different feature representations were created:

- Flattened feature vectors for Logistic Regression and MLP models.
- Original image tensors for CNN models.

Flattening converts each image into a one-dimensional vector, allowing traditional machine learning models and fully connected neural networks to process the data. However, this representation discards local spatial relationships between neighbouring pixels. In contrast, CNN models retain the original two-dimensional image structure so that convolutional filters can learn local texture and spatial patterns directly.

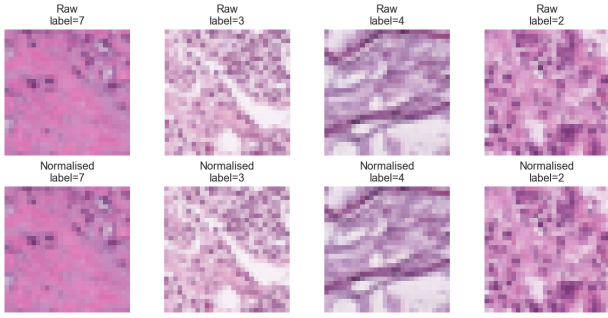


Fig. 4. Comparison between raw and normalised images.

For Logistic Regression and MLP, flattened features were additionally standardised using `StandardScaler`. Standardisation ensures that input features have comparable scales, which improves optimisation efficiency and prevents features with larger numeric ranges from dominating gradient updates. CNN models used only image normalisation because convolutional operations are generally less sensitive to feature scale differences and already exploit local spatial structure effectively.

Several additional preprocessing techniques were considered but ultimately not applied. Data augmentation was not included because the assignment focuses primarily on comparing model architectures rather than maximising benchmark performance. Dimensionality reduction methods such as PCA were also avoided because they may remove fine-grained local texture information that is important for histopathology classification.

Figure 4 compares raw and normalised images.

III. METHODS

A. Model Selection and Motivation

This study compares three classification approaches with different levels of modelling complexity: Logistic Regression, a multilayer perceptron (MLP), and a convolutional neural network (CNN). The models were intentionally selected to represent progressively more sophisticated approaches to image classification.

Logistic Regression was chosen as a traditional machine learning baseline from the first six weeks of the course. Despite its simplicity, it provides an interpretable reference point for evaluating whether more complex neural network architectures offer meaningful improvements on the PathMNIST dataset.

The MLP was included as an intermediate nonlinear model. Compared with Logistic Regression, the MLP can learn nonlinear feature interactions through hidden layers and activation functions. However, because it still operates on flattened image vectors, it does not explicitly preserve spatial image structure.

Finally, CNNs were selected because convolutional architectures are specifically designed for image data. Histopathology classification depends heavily on local tissue texture, spa-

tial organisation, and microscopic structural patterns, making CNNs particularly suitable for this task.

Comparing these three approaches allows analysis of the trade-offs between interpretability, computational cost, parameter complexity, and predictive performance.

B. Comparison of Strengths and Weaknesses

The three selected models represent different levels of modelling complexity and provide useful trade-offs between interpretability, computational efficiency, and predictive performance.

Logistic Regression is computationally efficient and highly interpretable because predictions are produced through linear combinations of input features. The model contains relatively few trainable parameters and trains quickly even on large datasets. However, its linear decision boundaries are insufficient for modelling the complex nonlinear visual structures present in histopathology images. In addition, flattening the images removes spatial relationships between neighbouring pixels, limiting the model’s ability to capture local tissue patterns.

The MLP introduces nonlinear representation learning through hidden layers and activation functions, allowing more flexible decision boundaries than Logistic Regression. This increases predictive capacity and improves classification performance. However, the MLP still operates on flattened image vectors, meaning that spatial image structure is not explicitly preserved. The larger number of parameters also increases the risk of overfitting and leads to longer training times.

CNNs are particularly well suited to image classification because convolutional filters learn local spatial features directly from image data. Parameter sharing and local receptive fields allow CNNs to model texture patterns and hierarchical visual structures more efficiently than fully connected networks. These properties are especially important for PathMNIST, where many tissue categories differ through subtle local texture variations rather than large global colour differences. However, CNNs are computationally more expensive, require substantially longer training times, and are generally less interpretable than simpler linear models.

Overall, the comparison highlights a clear trade-off between simplicity and predictive performance. Simpler models provide faster training and easier interpretability, while more complex neural network architectures achieve substantially stronger classification accuracy by learning richer feature representations.

C. Logistic Regression

For this dataset, Logistic Regression was selected as the machine learning algorithm for the first six weeks. Logistic regression features a simple structure, is easy to implement, and offers greater code interpretability compared to other machine learning algorithms. Additionally, it has fewer parameters and faster training speeds, making it an ideal baseline to compare against non-linear models utilizing neural networks. This comparison helps quantify the exact performance improvements achieved by more advanced models.

The implemented logistic regression model utilizes L_2 regularization, with the hyperparameter C controlling the regularization strength. During subsequent hyperparameter tuning, only a few representative values of C need to be evaluated, which further contributes to the model’s ease of implementation. However, because this algorithm is linear and requires flattening multi-dimensional features, its decision boundaries struggle to accurately classify the complex features present in the PathMNIST dataset. Consequently, from the initial design phase, its expected accuracy was anticipated to be significantly lower than that of neural network-based models.

Regarding the model architecture, it takes a flattened image vector ($28 \times 28 \times 3$) as input and outputs predictions for nine distinct classes. To optimize performance across different processing units, CPU execution employs the `lbfgs` solver with `max_iter` set to 5000 to ensure convergence. Conversely, GPU execution leverages the `qn` solver from the CuML library, along with an L_2 penalty, to accelerate both training and inference. For hyperparameter selection, the model’s simplicity dictates that C is the only hyperparameter required to tune the regularization strength. We selected a range of representative C values, spanning from:

$$C \in \{10^{-5}, 5 \times 10^{-5}, \dots, 5 \times 10^{-1}, 1\}$$

This comprised 11 candidate values in total. The model was trained using grid search, evaluating both accuracy and the macro F1 score on the validation set.

D. Multilayer Perceptron

The MLP extends Logistic Regression by introducing multiple fully connected hidden layers with nonlinear activation functions. The hidden layer transformation can be expressed as:

$$h = \sigma(Wx + b)$$

where x represents the input vector, W the weight matrix, b the bias vector, and σ the activation function.

Compared with Logistic Regression, the MLP can learn substantially more complex nonlinear relationships between input features and class labels. ReLU activation functions were used because they are computationally efficient and reduce vanishing gradient issues during optimisation. Dropout regularisation was also included to reduce overfitting by randomly deactivating neurons during training.

Although the MLP provides greater representational capacity, it still relies on flattened image vectors as input. Consequently, the model cannot explicitly preserve local spatial structure within the images. This limitation is important for histopathology classification because neighbouring pixel relationships often contain clinically meaningful texture information. The fully connected architecture also introduces a large number of trainable parameters, increasing both runtime and overfitting risk. The final MLP architecture selected after tuning is summarised in Table II. Each PathMNIST image was flattened from $28 \times 28 \times 3$ into a 2352-dimensional input

TABLE II
FINAL MLP ARCHITECTURE

Layer	Output Size
Input	2352
Dense (ReLU)	256
Dropout (0.3)	256
Dense (ReLU)	256
Dropout (0.3)	256
Dense (Softmax)	9

TABLE III
MLP HYPERPARAMETER SEARCH SPACE

Hyperparameter	Values
Hidden layers	{1, 2}
Hidden neurons	{64, 128, 256}
Dropout rate	{0.3, 0.4, 0.5}
Learning rate	{0.001, 0.0005, 0.0001}

vector. The model used two hidden dense layers with ReLU activation, dropout regularisation after each hidden layer, and a softmax output layer for nine-class classification.

The final MLP contained 670,473 trainable parameters.

1) *MLP Hyperparameter Tuning*: The MLP hyperparameter search focused on four key architectural and optimisation choices:

- Number of hidden layers
- Number of hidden neurons
- Dropout rate
- Adam learning rate

The number of hidden layers controls model depth and nonlinear representational capacity. Additional layers may improve feature learning but also increase optimisation difficulty and overfitting risk.

The number of hidden neurons determines the width of each layer and strongly affects model capacity. Larger hidden layers can represent more complex feature interactions but substantially increase parameter count.

Dropout rate controls the strength of regularisation. Higher dropout probabilities reduce overfitting by preventing co-adaptation between neurons, although excessive dropout may slow convergence and reduce effective model capacity.

The learning rate controls the optimisation step size used by the Adam optimiser. Larger learning rates accelerate convergence but may produce unstable optimisation, while smaller learning rates improve stability at the cost of longer training time.

An exhaustive grid search strategy was used because the search space remained computationally manageable while allowing systematic comparison across all parameter combinations.

This resulted in $2 \times 3 \times 3 \times 3 = 54$ total hyperparameter combinations. The search was performed using KerasTuner grid search with validation accuracy as the selection criterion.

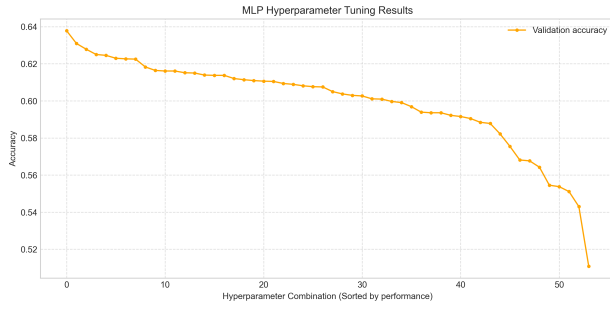


Fig. 5. MLP hyperparameter tuning results.

TABLE IV
FINAL CNN ARCHITECTURE

Layer	Output Shape
Conv2D (64 filters)	$28 \times 28 \times 64$
MaxPooling2D	$14 \times 14 \times 64$
Conv2D (128 filters)	$14 \times 14 \times 128$
MaxPooling2D	$7 \times 7 \times 128$
Flatten	6272
Dense (128 units)	128
Dropout (0.3)	128
Dense (Softmax)	9

E. Convolutional Neural Network

The CNN architecture was designed specifically to preserve spatial image structure during feature extraction. Unlike Logistic Regression and MLP models, CNNs process the original two-dimensional image tensors directly rather than flattened feature vectors.

The final CNN architecture contained:

- Two convolutional layers
- Two max-pooling layers
- One fully connected dense layer
- One dropout layer
- One softmax output layer

The convolutional layers used 3×3 kernels with ReLU activation and `same` padding. Small convolution kernels were selected because they efficiently capture local texture patterns while keeping parameter count relatively low. Max-pooling layers reduced spatial dimensionality and computational cost while improving translational robustness.

The final CNN architecture is summarised in Table IV.

CNNs provide several important advantages for this dataset. Convolutional filters preserve neighbouring pixel relationships and automatically learn hierarchical image features ranging from simple edges to more complex tissue structures. Parameter sharing also improves computational efficiency compared with fully connected architectures of similar representational power.

These advantages are particularly important for PathMNIST because many classes differ primarily through subtle local texture variations rather than large global colour differences. CNNs are therefore expected to perform substantially better than flattened-feature models.

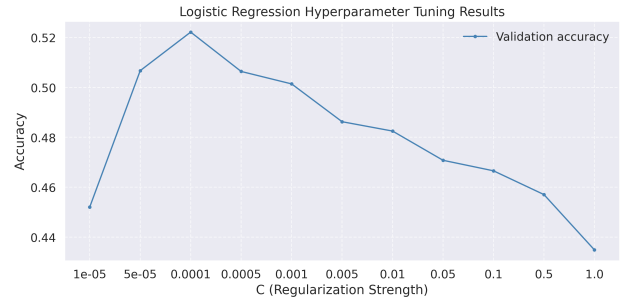


Fig. 6. LR hyperparameter tuning results.

However, CNNs also introduce greater computational complexity. Training requires longer runtime and more memory because convolution operations are repeatedly applied across the image. CNNs additionally contain many interacting hyperparameters, making optimisation and model selection more difficult. Their higher representational capacity also increases overfitting risk without appropriate regularisation.

1) *CNN Hyperparameter Tuning*: Four CNN hyperparameters were selected for exhaustive grid search:

- Number of convolutional filters
- Dense layer width
- Dropout rate
- Adam learning rate

The number of convolutional filters controls how many feature detectors are learned at each convolutional layer. Increasing filter count improves representational capacity but also increases parameter count and computational cost.

Dense layer width determines the complexity of the final classification stage after convolutional feature extraction. Larger dense layers may improve nonlinear decision boundaries but can also increase overfitting risk.

Dropout rate controls regularisation strength by randomly deactivating hidden neurons during training. Higher dropout values improve generalisation but may slow optimisation.

Learning rate determines the optimisation step size for Adam. Larger values accelerate training but may produce unstable convergence, while smaller values generally improve optimisation stability at the cost of slower learning.

The search space contained $2 \times 3 \times 3 \times 3 = 54$ combinations. Exhaustive grid search was selected because the total number of configurations remained computationally feasible while ensuring systematic comparison across all candidate hyperparameter settings.

The best-performing CNN configuration used:

- `conv_filters` = 64
- `dense_units` = 128
- `dropout_rate` = 0.3
- `learning_rate` = 0.001

The highest validation accuracy achieved during tuning was approximately 80.05%.

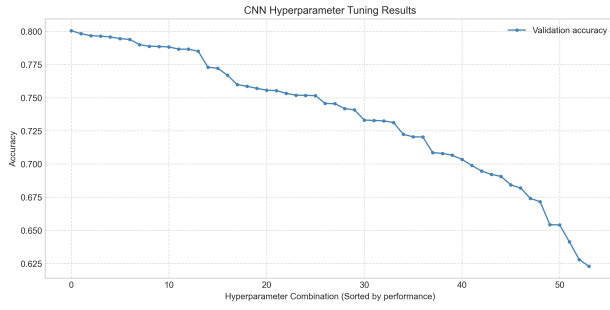


Fig. 7. CNN hyperparameter tuning results.

IV. RESULTS AND DISCUSSION

A. Hyperparameter Tuning Results

1) Results of Logistic Regression Hyperparameter Tuning:

It should be noted that due to the utilization of GPU-accelerated training and inference, the `qn` solver was employed. In this configuration, the accuracy cannot be perfectly fixed via random seeds, leading to minor fluctuations in performance metrics across different training runs. Consequently, the visualized data presented in this report may not be identical to results generated by subsequent executions of the code.

In the hyperparameter tuning phase, a grid search was conducted across a range of C values to identify the optimal configuration and prevent overfitting. As illustrated in the Figures 6, an excessively large C (lower regularization) leads to overfitting, resulting in a decline in validation accuracy. Conversely, a smaller C (higher regularization) allows the model to learn more generalized features. However, an overly small C can be detrimental, as it causes underfitting and subsequent performance degradation. Ultimately, the model achieved its peak performance on the validation set when $C = 10^{-4}$.

2) Results of MLP and CNN Hyperparameter Tuning:

Figures 5 and 7 summarise the validation accuracy obtained during hyperparameter tuning for the MLP and CNN models.

The tuning results show clear performance differences across hyperparameter combinations. For the MLP, increasing network width and depth generally improved validation accuracy because larger hidden representations allowed the model to capture more complex nonlinear feature relationships. However, the improvement became less significant for larger architectures, suggesting diminishing returns from additional parameters.

For the MLP, the best validation accuracy achieved during tuning was 63.78%. The selected configuration used two hidden layers, 256 hidden neurons per layer, a dropout rate of 0.3, and an Adam learning rate of 0.0005. This indicates that the MLP benefited from a deeper and wider architecture, but still required moderate dropout regularisation to reduce overfitting on the high-dimensional flattened image vectors.

The CNN tuning results exhibited a similar trend. Configurations with larger convolutional filter counts generally achieved higher validation accuracy because additional filters

increased the model’s ability to learn diverse local texture patterns from histopathology images. Moderate dropout values also produced more stable performance than very high dropout rates, indicating that excessive regularisation reduced effective model capacity.

The best-performing CNN configuration used 64 convolutional filters, 128 dense units, a dropout rate of 0.3, and a learning rate of 0.001, achieving a validation accuracy of approximately 80.05%. This result aligned closely with expectations prior to experimentation. Because PathMNIST classification depends heavily on local texture structure, architectures capable of preserving spatial information were expected to outperform flattened-feature approaches.

An important observation from the tuning process is that larger models did not always achieve better performance. Some high-capacity configurations increased runtime substantially while producing only marginal validation improvements. This suggests that moderate model complexity provided a better balance between representational power and generalisation for the relatively small 28×28 image resolution used in this dataset.

3) *Runtime of Hyperparameter Tuning:* Table V summarises the recorded runtime information for the hyperparameter tuning stage. Logistic Regression recorded the full tuning runtime directly in the notebook, while the MLP and CNN searches were reloaded from saved KerasTuner results in the exported notebook. Therefore, the displayed 0.00 minute runtime for MLP and CNN represents cache loading time rather than the original full search time.

The runtime behaviour is consistent with the theoretical complexity of the three model families. Logistic Regression was the simplest search because only the regularisation parameter C was varied, and each candidate required only one model fit. Even so, weaker regularisation values required longer fitting time, with large C values taking substantially longer than strongly regularised models. This suggests that less regularised linear models were more difficult to optimise on the high-dimensional flattened image features.

Although the original full runtime for the MLP and CNN grid searches was not preserved in the exported notebook because KerasTuner reloaded cached results, their computational cost was expected to be substantially higher than Logistic Regression. Each MLP and CNN tuning run involved 54 candidate configurations trained over multiple epochs. The MLP search was more expensive than Logistic Regression because each candidate architecture required repeated forward and backward passes through dense layers. The CNN search was expected to be the most computationally expensive because convolution operations were repeatedly applied across spatial feature maps for every candidate configuration.

This runtime pattern reinforces the practical trade-off observed in the final results: more complex neural architectures achieved stronger validation and test performance, but required greater computational cost during tuning and training.

TABLE V
AVAILABLE RUNTIME INFORMATION FOR HYPERPARAMETER TUNING

Model	Search Space	Training Setup	Best Validation Accuracy	Recorded Runtime
Logistic Regression	11 combinations	One fit per C value	0.5191	11.68 min
MLP	54 combinations	10 epochs per trial	0.6378	Cached reload in notebook
CNN	54 combinations	10 epochs per trial	0.8005	Cached reload in notebook

B. Final Model Comparison

Table VI compares the final test-set performance of all three models.

Utilizing the optimal hyperparameter ($C = 10^{-4}$) derived from the tuning phase, the Logistic Regression model was evaluated on the unseen test set, achieving an accuracy of approximately 52.74%. While this accuracy is relatively low when compared to other models, it effectively corroborates our previously stated hypothesis.

An analysis of the confusion matrix reveals that Logistic Regression demonstrates strong recognition capabilities for certain categories, specifically those corresponding to the high-value cells within the matrix. Conversely, its performance across several other classes remains suboptimal. As a linear machine learning model, Logistic Regression inherently requires flattened data as input, which inevitably leads to the loss of spatial information. Consequently, the model struggles to accurately classify categories that exhibit similar global color or brightness distributions but differ in fine-grained structural details. This highlights the fundamental limitations of simple models like Logistic Regression, and simultaneously underscores one of the primary motivations for developing neural networks to tackle such complex classification tasks.

The CNN achieved the strongest overall performance across all evaluation metrics, substantially outperforming both Logistic Regression and the MLP. This result is consistent with the theoretical advantages of convolutional architectures for image classification tasks. Convolutional filters preserve local spatial relationships and automatically learn hierarchical visual features, which are particularly important for histopathology image analysis.

The MLP achieved a noticeable improvement over Logistic Regression, reaching a test accuracy of 67.06%, macro F1-score of 0.665, and weighted F1-score of 0.669. This improvement shows that nonlinear hidden layers allowed the model to learn more flexible decision boundaries than the linear baseline. However, performance remained substantially lower than the CNN because the MLP still relied on flattened image vectors. Flattening removed local spatial relationships before classification, which limited the model’s ability to distinguish tissue categories that differ mainly through subtle texture and arrangement patterns rather than large global colour differences.

The runtime comparison highlights a clear trade-off between predictive performance and computational cost. Logistic Regression trained quickly and used relatively few parameters, while the CNN required substantially longer training time

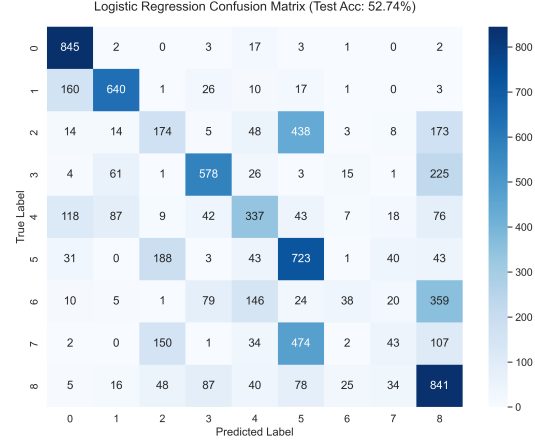


Fig. 8. Logistic Regression confusion matrix.

because of repeated convolution operations and significantly larger parameter complexity. However, the large improvement in predictive performance suggests that the additional computational cost was justified for this image classification task.

Another important observation is that the CNN achieved better parameter efficiency than might initially be expected. Although the CNN contained more trainable parameters overall, convolutional parameter sharing allowed substantially stronger feature extraction compared with fully connected architectures using flattened inputs. This demonstrates why CNNs remain highly effective for image-based learning problems even when computational cost increases.

The detailed class-level error patterns are further examined using confusion matrices in the next subsection.

C. Confusion Matrix Analysis

Figures 8, 9, and 10 show the confusion matrices for the three models.

The confusion matrices reveal several important differences between the models beyond overall accuracy values.

The Logistic Regression confusion matrix shows widespread off-diagonal errors across many classes. This suggests that the linear decision boundaries were insufficient for separating visually complex tissue categories. Misclassifications occurred particularly frequently between classes with similar colour distributions, indicating that the model relied heavily on global intensity patterns rather than detailed structural information.

The MLP confusion matrix demonstrated improved class separation compared with Logistic Regression, particularly

TABLE VI
FINAL MODEL COMPARISON

Model	Best Hyperparameters	Test Accuracy	Macro F1	Weighted F1	Runtime (s)	Parameters
Logistic Regression	$C = 10^{-4}$	0.527	0.468	0.484	38.40	21,177
MLP	hidden layers = 2, hidden neurons = 256, dropout = 0.3, learning rate = 0.0005	0.671	0.665	0.669	63.10	670,473
CNN	conv filters = 64, dense units = 128, dropout = 0.3, learning rate = 0.001	0.835	0.833	0.835	640.52	879,753

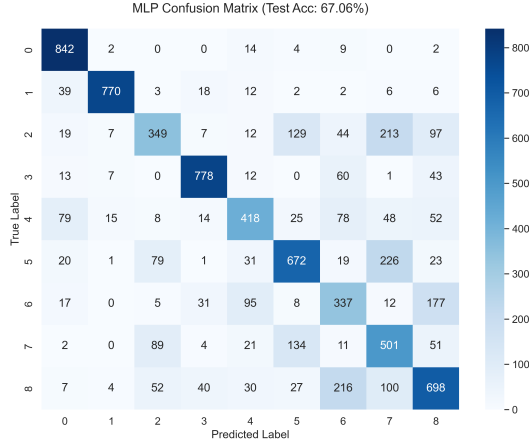


Fig. 9. MLP confusion matrix.

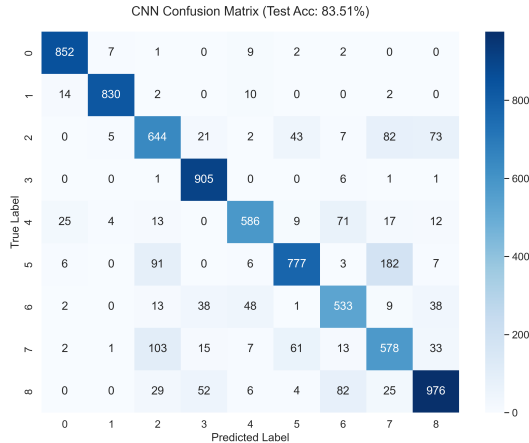


Fig. 10. CNN confusion matrix.

for visually distinctive classes. Classes 0, 1, and 3 achieved relatively strong F1-scores of approximately 0.881, 0.926, and 0.861 respectively, indicating that the MLP could identify some tissue categories reliably after learning nonlinear feature combinations. However, classes 2, 6, and 7 were more difficult, with F1-scores of approximately 0.477, 0.462, and 0.522. These weaker results suggest that the MLP struggled when classes shared similar staining patterns or subtle local texture differences. Although nonlinear hidden layers improved feature representation, the flattened image representation still

limited the model’s ability to learn local spatial relationships.

The CNN confusion matrix exhibited substantially stronger diagonal concentration, indicating more consistent classification across nearly all classes. This result suggests that convolutional feature extraction successfully captured discriminative local image patterns. Nevertheless, some confusion persisted between visually similar tissue classes. For example, several misclassifications occurred among classes with highly similar staining textures and microscopic structures. These errors are understandable because some PathMNIST categories remain difficult even for more advanced image models due to overlapping histopathological appearance.

Another important observation is that CNN performance improvements were not uniform across all classes. Certain categories achieved near-perfect recall, while others remained comparatively difficult. This indicates that class-specific visual similarity remains a significant challenge even when spatial structure is preserved.

D. Overall Discussion

Overall, the experimental results strongly support the hypothesis that preserving spatial structure is essential for histopathology image classification. Both Logistic Regression and the MLP relied on flattened image vectors, which removed relationships between neighbouring pixels before classification. In contrast, the CNN directly processed two-dimensional image tensors and learned hierarchical spatial representations through convolutional filters.

The results also demonstrate that increased model complexity alone does not automatically guarantee better performance. While the CNN substantially outperformed the simpler models, hyperparameter tuning showed that excessively large architectures often produced only small validation improvements while increasing runtime and overfitting risk. Effective model design therefore requires balancing representational power, computational cost, and generalisation performance.

From a practical perspective, the findings suggest that convolutional architectures are significantly more suitable for biomedical image classification tasks involving local texture information. However, simpler models may still remain useful in scenarios where computational resources, runtime constraints, or interpretability are more important than maximising predictive accuracy.

One limitation of this study is that all experiments were performed using relatively low-resolution 28×28 images.

Higher-resolution histopathology images may contain additional structural information that could further increase the performance gap between convolutional and non-convolutional models. Future work could therefore investigate deeper CNN architectures, data augmentation strategies, transfer learning approaches, or higher-resolution datasets.

V. CONCLUSION

This study investigated the performance of three classification approaches — Logistic Regression, a multilayer perceptron (MLP), and a convolutional neural network (CNN) — on the PathMNIST biomedical image classification task.

The experimental results demonstrated a clear relationship between model architecture and classification performance. Logistic Regression provided a simple and computationally efficient baseline but struggled to model the complex nonlinear visual patterns present in histopathology images. The MLP achieved improved performance by learning nonlinear feature representations; however, the flattening operation removed important spatial relationships between neighbouring pixels, limiting its ability to distinguish visually similar tissue categories. In contrast, the CNN consistently achieved the strongest performance across accuracy and F1-score metrics because convolutional layers preserved local spatial structure and learned hierarchical texture representations directly from image data.

The comparison also highlighted an important practical trade-off between predictive performance and computational cost. Although the CNN required substantially longer runtime and contained a larger number of trainable parameters, the improvement in classification performance was large enough to justify the additional complexity for this task. These findings suggest that preserving spatial information is critically important for biomedical image classification problems where subtle local texture differences determine class membership.

Several limitations should also be acknowledged. First, all experiments were conducted using relatively low-resolution 28×28 images, which may limit the amount of structural information available to the models. Second, the study focused on comparatively lightweight architectures because of computational constraints, meaning that more advanced CNN designs were not explored. Finally, evaluation was restricted to the PathMNIST benchmark dataset, so the results may not generalise directly to larger or more diverse histopathology datasets.

Future work could extend this study in several directions. One possible improvement would be to investigate deeper convolutional architectures or transfer learning methods using pretrained image models. Additional preprocessing and data augmentation techniques could also be explored to improve generalisation performance and reduce sensitivity to limited training variation. Furthermore, future studies could examine higher-resolution histopathology images and evaluate whether the observed advantages of convolutional architectures become even more significant when more detailed spatial information is available.

VI. REFLECTION

A. Student 540430668

During this assignment, the most important thing I learned was that building a machine learning model is not only about choosing a “more advanced” algorithm. Before starting the project, I expected the CNN to outperform the other models simply because it is specifically designed for image data. However, after implementing and tuning all three models, I realised that the difference in performance comes from much deeper reasons related to data representation, preprocessing, validation strategy, and model assumptions. For example, Logistic Regression and the MLP both relied on flattened image vectors, which removed the spatial relationships between neighbouring pixels. In contrast, the CNN was able to preserve local texture and structural information through convolutional filters, which became especially important for the PathMNIST dataset because many classes looked visually similar at a global level. This made me understand more concretely why CNNs are effective for image tasks, beyond just memorising the theory from lectures.

Another important lesson was how much practical machine learning depends on experimentation and careful evaluation rather than intuition alone. Some hyperparameter combinations that seemed “larger” or “more powerful” did not actually improve validation accuracy, and in some cases they increased runtime substantially without meaningful performance gains. I also realised how important it is to separate training, validation, and test data correctly to avoid biased model selection. During the implementation process, I spent a significant amount of time debugging environment issues, checking tensor shapes, interpreting confusion matrices, and comparing runtime trade-offs between models. This made the assignment feel much closer to a real machine learning workflow rather than a purely academic exercise. Overall, the assignment helped me better appreciate that successful machine learning involves both theoretical understanding and disciplined experimental practice.

B. Student 540776016

Through this assignment, the most important thing I learned was that the way image data is represented can strongly affect model performance. While implementing and analysing the MLP model, I initially expected that adding hidden layers and more neurons would be enough to achieve strong classification performance. The results showed that this was only partly true. The MLP clearly improved over Logistic Regression because it could learn nonlinear relationships between flattened pixel features, but it still performed much worse than the CNN. This helped me understand that model capacity alone is not sufficient for image classification if the input representation removes important spatial information.

I also learned the importance of systematic hyperparameter tuning. The MLP grid search showed that depth, width, dropout rate, and learning rate all influenced validation performance, and the best model was not simply the largest

or most aggressive configuration. The final MLP used two hidden layers with 256 neurons, moderate dropout, and a smaller learning rate, which provided a better balance between learning capacity and generalisation. This made me appreciate that machine learning experiments require careful validation and interpretation, rather than relying only on intuition about which model should be more powerful.

C. Student 520197321

Reflecting on the entire assignment, the most striking observation was the stark contrast between traditional machine learning algorithms and neural network models. Having implemented the Logistic Regression component, I clearly recognize its advantages: it requires tuning very few hyperparameters and is exceptionally straightforward to implement. While it is a highly efficient model, it is ill-suited for the complex PathMNIST dataset used in this assignment.

Most traditional machine learning models struggle to achieve satisfactory classification performance on such image datasets. This is primarily because models like Logistic Regression require multi-dimensional image tensors to be flattened into one-dimensional vectors, an operation that inherently destroys critical spatial topology and local feature relationships. Consequently, I believe Logistic Regression is far more appropriate for simpler, linearly separable tasks or tabular data, where it can deliver fast and highly accurate training and inference.

For complex datasets like PathMNIST, neural networks are significantly more capable of learning intricate, non-linear information. However, the choice of architecture within neural networks is equally critical. For instance, MLPs, much like Logistic Regression, require flattened inputs and therefore fail to precisely extract spatial features the way CNNs do with their convolutional kernels. Ultimately, this assignment reinforced the principle that model selection must be carefully deliberated and tightly aligned with the specific structural characteristics of the dataset and the task at hand.

D. Student 540255614

During this assignment, one of the most valuable things I learned was that model quality is not determined only by how many parameters a model has or how “advanced” it looks. After reading the final results carefully, I noticed that the difference between the MLP and the CNN was not simply a matter of model size. The MLP already had more than 670,000 parameters, which is not small at all, but its test accuracy was still only around 67%, while the CNN reached about 83.5%. This made me think more seriously about the importance of architecture. In this task, the main advantage of the CNN was not just higher complexity, but the fact that it could preserve local spatial structure and learn texture patterns directly from the image. For PathMNIST, where many classes have similar colours but subtle structural differences, this turned out to be much more important than just increasing the number of hidden units.

Another thing that stood out to me was that tuning results were often more informative than I expected. For example, Logistic Regression achieved its best validation accuracy at a very small value of C , and performance actually became worse as C increased, even while training time became much longer. This showed me that more flexibility does not always help, especially when the input is a high-dimensional flattened image vector. I also noticed that the best MLP settings were quite concentrated around deeper and wider configurations, but even with tuning, the improvement was still limited compared with the CNN. So for me, the most important lesson was that good machine learning work is not just about training models and reporting scores. It is about understanding why a model succeeds or fails, and linking that back to the structure of the data and the assumptions behind the method.

VII. AI ACKNOWLEDGEMENT

The primary generative AI tool used throughout this assignment was ChatGPT (OpenAI GPT-5). AI assistance was mainly used to help explain machine learning concepts, debug TensorFlow and Keras implementation issues, improve code readability, and refine the written presentation of the report.

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